

# Dominant Strategies and Nash Equilibrium

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## Iterated Elimination

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Today we'll look at games that cannot be solved with dominance alone.

We'll be using two tools in particular:

- iterated elimination and
- Nash equilibrium.

## Associated Reading

*Game Theory*, Sections 2.5 and 2.6

## When Dominance Alone Isn't Enough

Consider this game:

|   | L      | M      | R      |
|---|--------|--------|--------|
| T | (4, 3) | (2, 1) | (3, 0) |
| B | (2, 1) | (1, 2) | (1, 3) |

Player 2 (Column) has no dominated strategy. Each column does best against some row.

But we can still make progress by thinking about what rational players will rule out.

# Iterated Elimination of Dominated Strategies

The procedure:

1. Remove any strictly dominated strategy for any player.
2. In the smaller game that remains, look for newly dominated strategies.
3. Repeat until no more eliminations are possible.

This is justified by **common knowledge of rationality**: if everyone is rational and knows everyone is rational, no one will play a dominated strategy, so we can treat the game as if those strategies don't exist.

## Step 1: Check Player 1

|   | L      | M      | R      |
|---|--------|--------|--------|
| T | (4, 3) | (2, 1) | (3, 0) |
| B | (2, 1) | (1, 2) | (1, 3) |

Does T dominate B?

- Against L: T gives 4, B gives 2 → T better ☐
- Against M: T gives 2, B gives 1 → T better ☐
- Against R: T gives 3, B gives 1 → T better ☐

**T strictly dominates B.** Eliminate B.

## After Eliminating B

The game shrinks to one row:

|   | L      | M      | R      |
|---|--------|--------|--------|
| T | (4, 3) | (2, 1) | (3, 0) |

Now Player 2 chooses between L (payoff 3), M (payoff 1), R (payoff 0).

L strictly dominates M and R. Player 2 chooses L.

**Solution by iterated elimination:** (T, L) with payoffs (4, 3).

# The Limits of Iterated Elimination

Iterated elimination works neatly when it resolves the game completely. But often it doesn't.

|   | L      | C      | R      |
|---|--------|--------|--------|
| T | (3, 1) | (0, 2) | (1, 3) |
| M | (1, 3) | (1, 1) | (2, 0) |
| B | (0, 2) | (2, 0) | (3, 1) |

No row is dominated by another row, and no column is dominated by another column. Iterated elimination can't even get started here.

We need a solution concept that applies to any game.



## Nash Equilibrium

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## Best Response

A strategy is a **best response** to an opponent's strategy if no other strategy gives a strictly higher payoff against it.

If Column plays L, and Row's payoffs from T, M, B are 3, 1, 0, then T is Row's best response to L.

If the payoffs were 3, 3, 0, then both T and M would be best responses. "Best" means "tied for highest," not "uniquely highest." When using the bolding method, bold all tied-highest payoffs.

A rational player always plays a best response to whatever they expect the other player to do.

## Finding Best Responses

Go through each column and bold Row's highest payoff. Then go through each row and bold Column's highest payoff.

|                     | Ed: Normal | Ed: Extra |
|---------------------|------------|-----------|
| <b>Doug: Normal</b> | (2, 2)     | (0, 3)    |
| <b>Doug: Extra</b>  | (3, 0)     | (1, 1)    |

If Ed plays Normal: Doug gets 2 from Normal, 3 from Extra. Bold the 3.

If Ed plays Extra: Doug gets 0 from Normal, 1 from Extra. Bold the 1.

If Doug plays Normal: Ed gets 2 from Normal, 3 from Extra. Bold the 3.

If Doug plays Extra: Ed gets 0 from Normal, 1 from Extra. Bold the 1.

|              | Ed: Normal | Ed: Extra       |
|--------------|------------|-----------------|
| Doug: Normal | (2, 2)     | (0, <b>3</b> )  |
| Doug: Extra  | (3, 0)     | ( <b>1</b> , 1) |

## From Best Response to Nash Equilibrium

|              | Ed: Normal | Ed: Extra     |
|--------------|------------|---------------|
| Doug: Normal | (2, 2)     | (0, 3)        |
| Doug: Extra  | (3, 0)     | <b>(1, 1)</b> |

The cell where both payoffs are bolded is (Extra, Extra). Both players are playing a best response to each other. Neither would gain by switching.

A strategy profile is a **Nash equilibrium** if every player is playing a best response to what the others are doing.

**Definition:** A strategy profile  $(s_1, s_2, \dots, s_n)$  is a Nash equilibrium if for every player  $i$  and every alternative strategy  $s'_i$ :

$$\pi_i(s_i, s_{-i}) \geq \pi_i(s'_i, s_{-i})$$

# Nash Equilibrium in the Coordination Game

|             | Drive Left | Drive Right |
|-------------|------------|-------------|
| Drive Left  | (2, 2)     | (0, 0)      |
| Drive Right | (0, 0)     | (2, 2)      |

(Left, Left): given P2 plays Left, P1 gets 2 from Left vs 0 from Right. No improvement. ☐

(Right, Right): given P2 plays Right, P1 gets 2 from Right vs 0 from Left. No improvement. ☐

**Two Nash equilibria.** Each is stable; the problem is coordination.

## Nash Equilibrium in Battle of the Sexes

|          | Opera  | Football |
|----------|--------|----------|
| Opera    | (2, 1) | (0, 0)   |
| Football | (0, 0) | (1, 2)   |

(Opera, Opera): P1 gets 2; switch to Football gives 0. P2 gets 1; switch to Football gives 0.  $\boxed{?}$

(Football, Football): P1 gets 1; switch to Opera gives 0. P2 gets 2; switch to Opera gives 0.  $\boxed{?}$

Again two equilibria. This time each favors a different player.

# The Stag Hunt

Two hunters can cooperate to hunt a stag (high reward) or hunt a rabbit individually (safe reward).

|             | Hunt Stag | Hunt Rabbit |
|-------------|-----------|-------------|
| Hunt Stag   | (4, 4)    | (0, 3)      |
| Hunt Rabbit | (3, 0)    | (3, 3)      |

Let's find the Nash equilibria using the bolding method.



## Stag Hunt: Best Responses

Row's best responses: against Stag, Stag gives 4 vs Rabbit gives 3. Against Rabbit, Rabbit gives 3 vs Stag gives 0.

Column's best responses are identical (the game is symmetric).

|             | Hunt Stag     | Hunt Rabbit   |
|-------------|---------------|---------------|
| Hunt Stag   | <b>(4, 4)</b> | (0, 3)        |
| Hunt Rabbit | (3, 0)        | <b>(3, 3)</b> |

Two cells with both payoffs bolded: (Stag, Stag) and (Rabbit, Rabbit).

## Stag Hunt: Two Equilibria, Different Characters

|             | Hunt Stag | Hunt Rabbit |
|-------------|-----------|-------------|
| Hunt Stag   | (4, 4)    | (0, 3)      |
| Hunt Rabbit | (3, 0)    | (3, 3)      |

(Stag, Stag) is **Pareto superior**. Both do better than in (Rabbit, Rabbit).

But (Stag, Stag) is **riskier**: if your partner defects to Rabbit, you get 0.

(Rabbit, Rabbit) is **safer**: you get 3 no matter what the other player does.

## Prisoner's Dilemma vs. Stag Hunt

Both games have a Pareto-superior cooperative outcome that's not always achieved.

But they're structurally different:

- In the **PD**, defection is a *dominant* strategy. Rational even if you know the other will cooperate.
- In the **Stag Hunt**, cooperation is rational *if you trust* the other player will cooperate. But unilaterally it's risky.
- In PD the problem is incentives. In Stag Hunt the problem is **trust and coordination**.

What are some cases in your life that feel like PDs?

What are some cases that feel like Stag Hunts?

# What Nash Equilibrium Does and Doesn't Tell Us

Nash equilibrium identifies **stable** outcomes: situations where no one has reason to deviate.

It does **not** tell us:

- Which equilibrium will be reached when there are multiple.
- That equilibria are *good*. The PD equilibrium is Pareto inferior.
- How players decide what to do when they haven't communicated.

Nash equilibrium is a *necessary condition* for a stable rational outcome, not a complete theory of rational play.

## Three Kinds of Game Theory

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Bonanno's book focuses on **rational choice game theory**: assume common knowledge of rationality, deduce what rational players will do.

But game theory is also done in two other ways.

**Evolutionary game theory:** strategies are hard-wired, and populations of strategies grow proportionally to their success. No assumption of rationality. Selection does the work.

**Experimental game theory:** put real people in labs and see what they actually do. Results often diverge from rational-choice predictions.



## Why the Distinctions Matter

In the **PD**, rational-choice theory predicts (Defect, Defect).

Experimental evidence: people frequently cooperate, especially when they interact repeatedly. The Axelrod tournaments showed that “tit for tat” strategies do very well.

Evolutionary simulations: populations can evolve toward cooperation even without any rationality assumption.

All three approaches study the same games but ask different questions and reach different conclusions.

## Worked Examples

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## A Location Game

A seafront promenade has seven numbered stalls (1 through 7, left to right). Two ice cream vendors choose a stall simultaneously. Beachgoers buy from the nearer vendor; ties split equally. There are 14 customers, 2 at each stall.

Where should a rational vendor locate?

## Position 4 Beats Position 1

Compare market share when you play **1** vs. **4**, for each possible opponent location:

| Opponent at | Your share at 1 | Your share at 4 |
|-------------|-----------------|-----------------|
| 1           | 7               | 10              |
| 2           | 2               | 9               |
| 3           | 3               | 8               |
| 4           | 4               | 7               |
| 5           | 5               | 8               |
| 6           | 6               | 9               |
| 7           | 7               | 10              |

**Position 4 strictly dominates position 1.** By symmetry it also strictly dominates position 7.

## Round 1: Delete 1 and 7

No rational vendor will play 1 or 7. The effective game shrinks to positions  $\{2, 3, 4, 5, 6\}$ .

Now apply the same reasoning within the reduced game.

## Position 4 Beats Position 2 (in the Reduced Game)

| Opponent at | Your share at 2 | Your share at 4 |
|-------------|-----------------|-----------------|
| 2           | 7               | 9               |
| 3           | 4               | 8               |
| 4           | 5               | 7               |
| 5           | 6               | 8               |
| 6           | 7               | 9               |

**Position 4 strictly dominates position 2** in the reduced game, and by symmetry dominates position 6.

## Round 2: Delete 2 and 6

The game shrinks further to  $\{3, 4, 5\}$ . Apply the reasoning once more.

## Position 4 Beats Position 3 (in the Reduced Game)

| Opponent at | Your share at 3 | Your share at 4 |
|-------------|-----------------|-----------------|
| 3           | 7               | 8               |
| 4           | 6               | 7               |
| 5           | 7               | 8               |

**Position 4 strictly dominates position 3** in this reduced game, and by symmetry dominates position 5.



## Round 3: Delete 3 and 5

Only position 4 survives. Both vendors locate at the centre.

This generalises: on any symmetric distribution of customers, rational vendors converge on the **median** location. The same logic applies to political candidates, petrol stations, fast food chains, and any competitor serving a distributed market.

Position 4 is also the unique Nash equilibrium of this game. Neither vendor can do better by moving, given the other is at the centre.

An airline loses two passengers' identical antiques. It asks each to independently claim a value between \$200 and \$600. Both receive the *lower* claim, with adjustments:

- The lower claimer gets a \$200 bonus.
- The higher claimer gets a \$200 penalty.

Equal claims pay that amount to both.

## The Case Against Claiming \$600

If your opponent claims \$600 and you claim \$500, you receive  $500 + 200 = \$700$ , beating the \$600 you'd get by matching them.

So against an opponent who claims \$600, claiming \$500 is better.

But if the opponent claims \$400 or less, both \$500 and \$600 yield the same payoff (you both get the opponent's amount minus \$200). So \$500 only *weakly* dominates \$600, not strictly.

Suppose rational players won't claim \$600. The game reduces to  $\{200, 300, 400, 500\}$ . Now \$500 is weakly dominated by \$400 in the reduced game, by the same argument.

Each round knocks out the top claim. The only Nash equilibrium is for both to claim \$200.

## The Problem with That Prediction

In experiments, most participants claim close to \$600. Average claims typically sit around \$500. Almost nobody claims \$200.

The step-by-step argument is valid at each stage, yet the conclusion is both worse for everyone and empirically rare.

This is one reason game theorists take experimental evidence seriously. Common knowledge of rationality is doing a lot of work in the iterated argument, and real players seem to stop the iteration early, or factor in that their opponents will too.

- We move from simultaneous games to **dynamic games**, where players move in sequence.
- This requires a new representation: the **game tree** (extensive form).
- And a new solution concept: **backward induction**.